Lightweight CNN Crack Detection on Rail Heads

Abstract.

We test whether a lightweight convolutional neural network (CNN) running on a single-board computer can find surface cracks on rail heads from camera images. On our test set the network found 91% of cracked images in daylight.

Method.

A downward-facing camera on a push trolley photographs the rail head every 5 cm. Each 224 by 224 pixel image is classified as cracked or intact by a MobileNet-style CNN with 1.2 million parameters. The model runs on a Raspberry Pi 4 at 14 frames per second, so no cloud connection is needed.

Data.

We collected 2,400 rail-head images on a 2 km test line, of which 600 show cracks wider than 0.5 mm. The images were split 70/15/15 into training, validation and test sets.

Note.

A fictional sample paper by A. Rao and M. Chen (2024), written to test MAIROS. It is not published research: its data and results are invented.

Results.

On the daylight test set the CNN reached a recall of 0.91 and a precision of 0.88. On 300 images taken at dusk, recall fell to 0.64. Cracks narrower than 0.5 mm were found in only 22% of cases.

Limitations.

A limitation of this study is that all training images were captured in daylight, so performance at night or in rain is unknown. The trolley was pushed at walking speed, and motion blur at higher speeds was not measured.

Open questions.

It remains unclear whether infrared illumination can restore recall at night. Future work should pair the camera with a second sensor that does not depend on light.

Conclusion.

A small CNN on a single-board computer can flag visible rail-head cracks in daylight without a cloud connection.